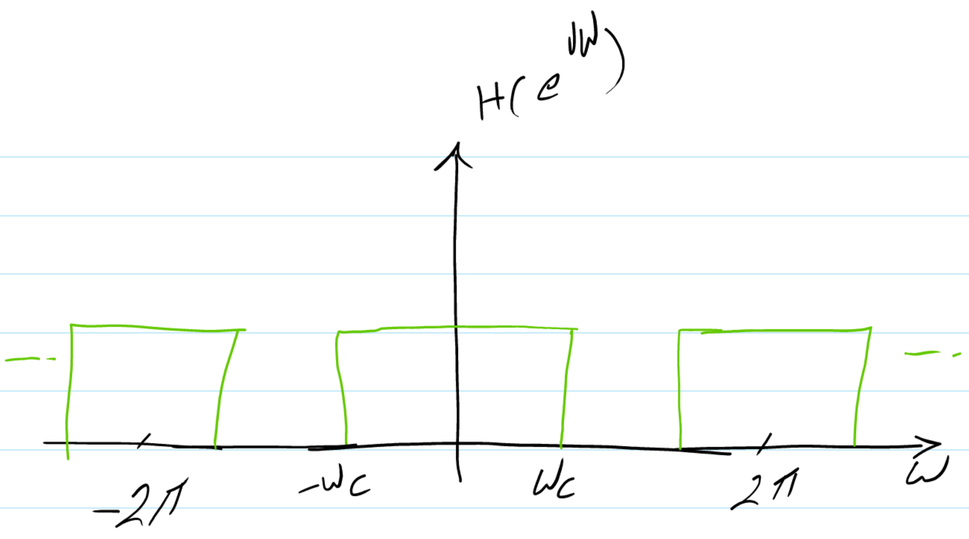
Lecture # 7

28 April 2026 07:19

Ideal low pass filter :-



* $H(e^{jw}) = \begin{cases} 1, & |w| < w_c \\ 0, & o.w \end{cases}$

$$h[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} H(e^{jw})\, e^{jwn}\, dw = \frac{1}{2\pi}\int_{-w_c}^{w_c} e^{jwn}\, dw = \frac{1}{2\pi}\left[\frac{e^{jwn}}{jn}\right]_{-w_c}^{w_c} = \frac{}{2\pi jn}$$

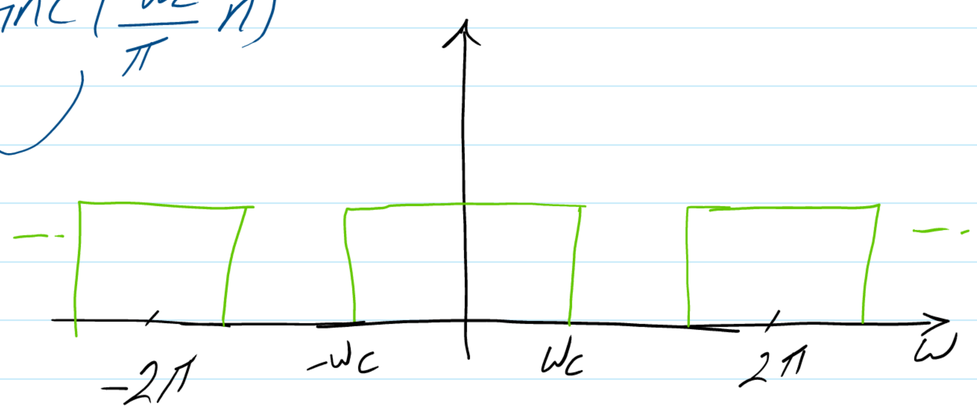
Trancated ideal filter :-

$h[n] = h[n] \cdot w[n]$

Ideal ↙ ↘ window

Ideal low pass filter : $h[n] = \frac{w_c}{\pi}\, sinc\left(\frac{w_c}{\pi} n\right)$

$sinc(x) = \frac{\sin(\pi x)}{\pi x}$



$w[n]$ →
- → rectangle
- → hanning
- → hamming
- → blackman
- → kaiser
- → triangle

the Problem with trancate is
- in Time domain it becomes like sinc function
- in frequency domain, it appears as we doing convolution with lowpass function.

filters
- Finite impulse Response (FIR)
- Infinite impulse response (IIR)

FIR filters :- like the trancated ideal filter

$h[n] = h[n] \cdot w[n]$

filteration is complitely through the convolution

$y[n] = x[n] * h[n]$

the I/o relation :

$$y[n] = a_0 x[n] + a_1 x[n-1] + \dots + a_m x_n[n-m] \rightarrow \text{in time Domain}$$

$$Y(e^{jw}) = (a_0 + a_1 e^{-jw} + \dots + a_m e^{-jmw}) \cdot X(e^{jw})$$

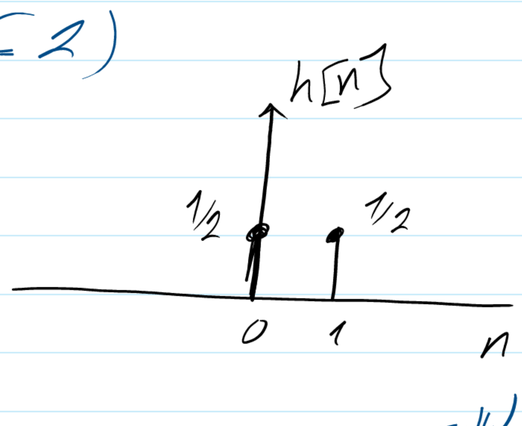
$$H(e^{jw}) = \frac{Y(e^{jw})}{X(e^{jw})} = a_0 + a_1 e^{-jw} + \dots + a_m e^{-jmw} \rightarrow \text{in freq Domain}$$

Example : $y[n] = \frac{1}{2}(x[n] + x[n-1])$, find $h[n]$.

(it's moving average filter order = 2)

solution :

$h[n] = \frac{1}{2}(\delta[n] + \delta[n-1])$

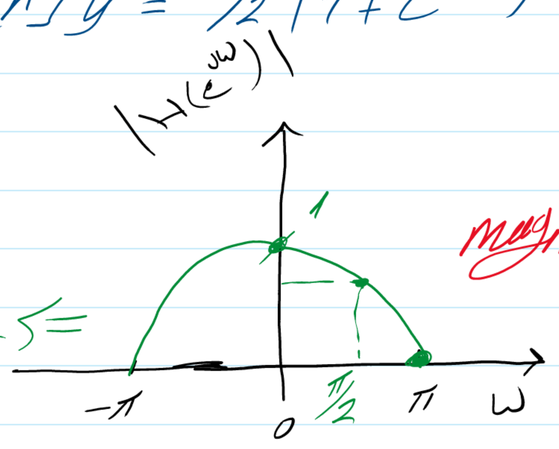


method 1 :

$$H(e^{jw}) = DTFT\{h[n]\} = \frac{1}{2}(1 + e^{-jw}) = \frac{e^{-jw}\left(e^{jw/2} + e^{-jw/2}\right)}{2} = \frac{2e^{-jw/2}\cos w/2}{2} =$$

$H(e^{jw}) = e^{-jw/2}\cos w/2$

Non Ideal Low Pass filter ≤=



Note

$e^{j\theta} = \cos\theta + j\sin\theta$

$e^{-j\theta} = \cos\theta - j\sin\theta$